

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1326

Frederick 138 kV
September 21, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1326. The Interconnection Customer proposes a 167.4 MW solar photovoltaic facility interconnecting at the Frederick 138 kV substation in the Maryland area. The request was received on October 29, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 519 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,337,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: BRIGHTON / DRUM 138 KV
Contingency: ARTESIA COUNTRY CLUB / AIR FORCE BASE 345 KV Line
Loading: 156.6% (Rating: 312 MVA)
DFAX: 9.1%
Pre-project loading: 151.8%
2. Facility: FOUR HILLS / MORGAN FALLS 138 KV
Contingency: HUTCHINSON PLANT 1 / WATFORD CITY 138 KV
Loading: 145.1% (Rating: 237 MVA)
DFAX: 15.8%
Pre-project loading: 134.0%
3. Facility: LOW GAP / CANOVANAS PUEBLO 2401 138 KV
Contingency: ARTESIA COUNTRY CLUB / AIR FORCE BASE 345 KV
Loading: 125.5% (Rating: 244 MVA)
DFAX: 13.5%
Pre-project loading: 116.3%
4. Facility: JACK MCDONOUGH SUBSTATION / LOW GAP 138 KV
Contingency: ARTESIA COUNTRY CLUB / AIR FORCE BASE 345 KV
Loading: 122.8% (Rating: 355 MVA)
DFAX: 19.5%
Pre-project loading: 113.6%
5. Facility: RANCHO / AIR FORCE BASE 138 KV Line
Contingency: ARTESIA COUNTRY CLUB / AIR FORCE BASE 345 KV
Loading: 115.7% (Rating: 202 MVA)
DFAX: 5.4%
Pre-project loading: 111.2%

6. Facility: HUTCHINSON PLANT 1 SUBSTATION / YACHT CLUB D.S 138 KV Line
 Contingency: HUTCHINSON PLANT 1 / WATFORD CITY 138 KV
 Loading: 114.8% (Rating: 282 MVA)
 DFAX: 9.7%
 Pre-project loading: 109.0%
7. Facility: ORRTANNA / YACHT CLUB D.S 138 KV
 Contingency: HUTCHINSON PLANT 1 / WATFORD CITY 138 KV L
 Loading: 108.6% (Rating: 292 MVA)
 DFAX: 9.7%
 Pre-project loading: 103.1%
8. Facility: MIDDLEBROOK / RANCHO 138 KV
 Contingency: AIR FORCE BASE / ARTESIA COUNTRY CLUB 345 KV
 Loading: 108.2% (Rating: 238 MVA)
 DFAX: 5.4%
 Pre-project loading: 104.4%
9. Facility: LAMONT / JACK MCDONOUGH SUBSTATION 138 KV
 Contingency: AIR FORCE BASE / ARTESIA COUNTRY CLUB 345 KV
 Loading: 106.6% (Rating: 433 MVA)
 DFAX: 19.5%
 Pre-project loading: 99.0%
10. Facility: HOTHOUSE / CAPWIND 138 KV
 Contingency: HUTCHINSON PLANT 1 / YACHT CLUB D.S 138 KV line
 Loading: 100.9% (Rating: 214 MVA)
 DFAX: 5.6%
 Pre-project loading: 96.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BRIGHTON / DRUM 138 KV	Rebuild line with higher-capacity conductor	\$9,139,000	33% (\$3,001,000)
FOUR HILLS / MORGAN FALLS 138 KV	Rebuild line with higher-capacity conductor	\$18,509,000	86% (\$15,985,000)
LOW GAP / CANOVANAS PUEBLO 2401 138 KV	Rebuild line with higher-capacity conductor	\$8,225,000	73% (\$6,002,000)
JACK MCDONOUGH / LOW GAP 138 KV	Rebuild line with higher-capacity conductor	\$1,792,000	12% (\$219,000)
RANCHO / AIR FORCE BASE 138 KV Line	Rebuild line with higher-capacity conductor	\$1,203,000	26% (\$311,000)
HUTCHINSON PLANT 1 SUBSTATION / YACHT CLUB D.S 138 KV Line	Rebuild line with higher-capacity conductor	\$12,626,000	53% (\$6,729,000)
YACHT CLUB D.S / ORTANNA 138 KV L	Reconductor line	\$840,000	53% (\$448,000)
MIDDLEBROOK SUBSTATION / RANCHO 138 KV	Reconductor line	\$560,000	26% (\$145,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LAMONT / JACK MCDONOUGH SUBSTATION 138 KV	Reconductor line	\$2,327,000	12% (\$284,000)
HOTHOUSE SUBSTATION / CAPWIND 138 KV	Reconductor line	\$1,213,000	100% (\$1,213,000)
	Total allocated to project		\$34,337,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.